

Functional Annotation Report: PP_3199 (UniProt Q88I03)

Pseudomonas putida KT2440 — a FAD-dependent flavonoid (naringenin) C-8 monooxygenase (FdeE ortholog)

1. Summary / Answer to the Research Question

PP_3199 (Q88I03) is a soluble, cytoplasmic, single-component FAD-dependent aromatic monooxygenase that catalyzes the committed first step of aerobic flavonoid (flavanone) catabolism in *Pseudomonas putida* KT2440. It is the ortholog (41.4 % full-length identity, conserved synteny) of the experimentally characterized **FdeE** enzyme of *Herbaspirillum seropedicae* SmR1, a **flavonoid C-8 hydroxylase (F8H)**. The enzyme uses FAD and NAD(P)H to activate molecular O₂ and inserts one oxygen atom as a hydroxyl group at the **C-8 position of the flavonoid A-ring**, converting **naringenin → 8-hydroxynaringenin (5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one)**. This introduces an *ortho*-dihydroxy (catechol/pyrogallol-type) A-ring that is the substrate for the adjacent ring-cleaving dioxygenase FdeC, funneling plant-derived flavonoids into central metabolism (ultimately oxaloacetate → TCA cycle). The enzyme has **broad substrate tolerance** across flavonoid subclasses (flavanones, flavones, isoflavones, flavonols). Its gene sits in a **flavonoid-inducible fde operon** controlled by a NodD-like LysR-type regulator (FdeR; KT2440 ortholog PP_3196), tying its function to a plant-associated/rhizosphere lifestyle.

Identity verification: The UniProt record (SubName "Monooxygenase", *P. putida* KT2440, OrderedLocusNames PP_3199, Pfam FAD_binding_3/PF01494; InterPro IPR002938, IPR050493, IPR036188) is fully consistent with all evidence gathered here. There is **no ambiguity** — PP_3199/Q88I03 corresponds to a specific FAD-dependent monooxygenase in the KT2440 flavonoid-degradation gene cluster. The direct enzymatic characterization derives from the *Herbaspirillum* ortholog; the KT2440 assignment rests on strong orthology, conserved gene-cluster synteny, and a conserved regulator (see Limitations).

2. Molecular Identity and Domain Architecture

Property	Value	Source
UniProt	Q88I03 (unreviewed/TrEMBL)	UniProtKB
Gene / locus	PP_3199 (GenBank AAN68807)	UniProt/KEGG
Organism	<i>P. putida</i> KT2440	UniProt
Length	374 aa	UniProt
Genome position	chr 3,628,211–3,629,335 (+ strand)	KEGG ppu:PP_3199
Pfam	PF01494 (FAD_binding_3) ×2	UniProt/KEGG
InterPro	IPR002938 (FAD-bd); IPR036188 (FAD/NAD-binding superfamily); IPR050493 (FAD-dep_Monooxygenase_BioMet)	UniProt
Fold (Gene3D/ SUPFAM)	3.50.50.60 / SSF51905 — FAD/NAD(P)-binding (Rossmann) domain	UniProt
PANTHER	PTHR13789 (MONOOXYGENASE)	UniProt
GO (inferred)	FAD binding (GO:0071949); monooxygenase activity (GO:0004497)	InterPro/ UniProt-KW

Class A single-component flavoprotein aromatic hydroxylase (p-hydroxybenzoate-hydroxylase, PHBH, family). The sequence carries the three diagnostic conserved fingerprints of this family:

1. **N-terminal Rossmann dinucleotide-binding motif GxGxxG** at residues 11–16 (GGGIGG) — binds the ADP moiety of FAD.
2. **Internal "GD/DG" fingerprint GADG** at residue 157 — part of the FAD/substrate-binding core.
3. **C-terminal "GD" motif GDAAH** at residue 287 — the hallmark motif of PHBH-family aromatic hydroxylases.

These features unambiguously place PP_3199 among the **Group A flavoprotein monooxygenases**, which are self-sufficient (single-polypeptide) enzymes that bind one tightly associated FAD, use NAD(P)H to reduce the flavin, and react the reduced flavin with O₂ to form a **C4a-(hydro)peroxyflavin** that performs electrophilic aromatic hydroxylation of an activated (typically phenolic) substrate, releasing the second O atom as water.

3. Genomic Context: the **fde** (flavonoid degradation) Cluster

PP_3199 is embedded in a syntenic flavonoid-catabolic gene cluster in KT2440 (KEGG orthology in parentheses):

Locus	Annotation	KEGG KO / Fde name
PP_3195	conserved protein	K26184 — FdeB
PP_3196	Nodulation protein d1 (LysR_substrate + HTH_1)	FdeR-type NodD/LysR regulator
PP_3197	Glyoxalase-family (VOC-superfamily) protein	K26181 — FdeC (ring-cleaving dioxygenase)
PP_3198	2Fe-2S ferredoxin	K26179 — FdeD
PP_3199	putative Monooxygenase	FdeE ortholog (K26180 unassigned in KEGG)
PP_3200	exported protein of unknown function	—
PP_3204	—	K26182 — FdeH
PP_3205	—	K26185 — FdeI
PP_3206	—	K26183 — FdeJ

The cluster's monooxygenase KO (FdeE, K26180) was not auto-assigned to any KT2440 gene by KEGG, yet PP_3199 is the **only** monooxygenase in the cluster and is the direct sequence ortholog of *Herbaspirillum* FdeE. The presence of the divergently transcribed **NodD-like LysR regulator PP_3196** mirrors the *Herbaspirillum* **fdeR** arrangement, showing the whole flavonoid-responsive regulatory + catabolic module is conserved.

4. Primary Function: Reaction, Substrate Specificity, and Regioselectivity

4.1 Reaction catalyzed

Naringenin + NAD(P)H + O₂ → 8-hydroxynaringenin (5,7,8-trihydroxyflavanone) + NAD(P)⁺ + H₂O.

Genetic and metabolite evidence in *H. seropedicae*: a Tn5 insertion in `fdeE` abolished growth on naringenin, and LC-MS/MS showed **naringenin remained unaltered** in the *fdeE* mutant, establishing FdeE as the enzyme performing the **initial step**. In the wild type, naringenin is "first mono-oxygenated by the FdeE protein, to produce 5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one, that is subsequently dioxygenated and cleaved at the A-ring by the FdeC dioxygenase" (Marin et al. 2016,).

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4.2 Regioselectivity and substrate range (biochemical)

Purified FdeE was directly characterized as a **flavin-dependent monooxygenase (FMO) that regioselectively introduces a hydroxyl at C-8** and is "active towards a wide range of flavonoids—flavanones, flavones, isoflavones, and flavonols"; it clusters with other **flavonoid-C8-hydroxylase (F8H)** FMOs, and was used preparatively to make 8-hydroxyquercetin (gossypetin) from quercetin (Dulak et al. 2024,).

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Interpretation: PP_3199's catalytic chemistry is **aromatic C–H hydroxylation at the C-8 carbon of the flavonoid A-ring**. On naringenin (A-ring = 5,7-dihydroxy resorcinol type), C-8 hydroxylation yields the 7,8-*ortho*-diol (a catechol), the electronic prerequisite for the subsequent **extradiol/meta ring cleavage** by the VOC-superfamily dioxygenase FdeC. Substrate specificity is **broad within the flavonoid scaffold** but scaffold-restricted (a "peripheral" aromatic-catabolism hydroxylase rather than a generalist).

4.3 Cofactors / accessory proteins

- **FAD** (bound; PF01494) and **NAD(P)H** (electron donor) are required for Group-A FMO turnover.

- The clustered **2Fe-2S ferredoxin FdeD (PP_3198)** and the flavin/redox partners encoded in the operon may support electron supply/flavin reduction and downstream redox chemistry of the pathway.

5. Biological Process and Pathway

PP_3199 initiates a **peripheral aromatic-catabolic (funneling) pathway** for plant flavonoids:

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Naringenin (plant flavanone)
|   PP_3199 (FdeE, FAD-dependent C-8 monooxygenase)  + O2, NAD(P)H
▼
8-Hydroxynaringenin (5,7,8-trihydroxyflavanone; A-ring catechol)
|   FdeC (PP_3197, VOC/glyoxalase-superfamily dioxygenase)  A-ring meta-cleavage
▼
ring-opened intermediates → ... (FdeB/D/H/I/J) ...
▼
oxaloacetate → TCA cycle (carbon/energy)

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The downstream steps and final production of oxaloacetate feeding the TCA cycle were defined for the *Herbaspirillum* pathway (Marin et al. 2016, [P 27059806](#)).

Thus PP_3199 enables the bacterium to **use plant-exuded flavonoids as carbon/energy sources** (and/or to detoxify them).

Regulation

The `fde` operon is a **flavonoid-inducible catabolic operon**. In *H. seropedicae*, transcription is activated by flavonoids (naringenin, chrysin, apigenin, luteolin) via the **NodD-like LysR-type regulator FdeR**, which is divergently transcribed from the operon (Wassem et al. 2017, [P 27878922](#));

mutating `fdeR` (or the first structural gene `fdeA`) abolishes naringenin degradation (Marin et al. 2013, [P 23125118](#)).

KT2440 conserves this architecture (regulator **PP_3196**, "Nodulation protein d1", LysR family).

Consequently PP_3199 is expressed **on demand** when flavonoids are encountered — consistent with a rhizosphere/plant-associated ecological role (flavonoids are major root exudates and rhizobial nod-gene inducers).

6. Subcellular Localization

Cytoplasmic. PP_3199 has no signal peptide and no predicted transmembrane segments, and Group-A flavoprotein hydroxylases (e.g., p-hydroxybenzoate hydroxylase) are classic **soluble cytoplasmic enzymes**. A Kyte–Doolittle scan (window 19) gives a maximum hydropathy of only **1.72** (below the ~1.9 transmembrane threshold), and this single mildly hydrophobic stretch (res 4–22) **coincides exactly with the N-terminal GxGxxG FAD-binding Rossmann motif** — a buried structural element, *not* a membrane anchor or signal peptide. The overall **GRAVY** = **0.011** is typical of a soluble globular protein. Flavonoid catabolism therefore proceeds intracellularly after uptake; a clustered putative transporter (PP_3202) and exported protein (PP_3200) may participate in substrate handling.

7. Evidence Summary

Claim	Evidence type	Strength
Class A FAD-dependent aromatic hydroxylase; FAD + NAD(P)H; C4a-peroxyflavin mechanism	Domain architecture (PF01494; IPR002938/036188/050493) + 3 conserved sequence fingerprints (GxGxxG, GADG, GDAAH)	Strong (bioinformatic)
Ortholog of characterized FdeE	Needleman-Wunsch global alignment: 41.4 % identity, 379 aligned positions vs Hsero_1007/D8J0W9; conserved synteny	Strong
Catalyzes naringenin → 8-hydroxynaringenin (first, committed step)	Genetics + LC-MS/MS in <i>H. seropedicae</i> (P 27059806)	Strong (ortholog, experimental)
Regioselective C-8 flavonoid hydroxylase (F8H), broad flavonoid range	In vitro biochemistry of purified FdeE (P 39107441)	Strong (ortholog, experimental)
Product feeds A-ring dioxygenase FdeC → TCA (oxaloacetate)	Metabolite profiling (P 27059806)	Strong (ortholog)
Flavonoid-inducible operon via NodD-like LysR FdeR	Genetics/transcription (P 27878922); (P 23125118) KT2440 PP_3196 conserved	Strong (ortholog) + genomic (KT2440)
Cytoplasmic localization	Sequence (no signal/TM) + family precedent	Moderate–strong (inference)

8. Supported and Refuted Hypotheses

Supported - H1: PP_3199 is a single-component FAD-dependent aromatic (phenolic) hydroxylase of the PHBH/Group-A family. ✓ (domains + motifs) - H2: Its physiological substrate is a plant flavonoid (naringenin), which it hydroxylates at C-8 as the committed step of flavonoid catabolism. ✓ (orthology + experimental data on FdeE) - H3: It functions in the cytoplasm within a flavonoid-inducible operon regulated by a NodD-like LysR regulator. ✓ (conserved synteny + regulator PP_3196)

Refuted / excluded - PP_3199 is *not* a two-component (flavin-reductase-dependent) monooxygenase system requirement mis-annotation, nor a generic "unknown monooxygenase": the specific F8H/PHBH-family fingerprints and cluster context define a precise reaction. - It is *not* a membrane or secreted protein (no signal peptide/TM).

9. Limitations and Future Directions

- **Direct KT2440 data are lacking:** all enzymatic/genetic characterization comes from the *H. seropedicae* ortholog (41.4 % identity) plus KT2440 genomic context. Direct heterologous expression/kinetics of PP_3199 itself, and a KT2440 PP_3199 knockout tested for growth on naringenin/flavones/isoflavones/flavonols, would confirm the assignment and quantify substrate preference (Km/kcat) and cofactor (NADH vs NADPH) selectivity.
 - **Exact downstream enzymology** (roles of FdeB/D/H/I/J, ring-opened intermediates) remains partly inferred.
 - **Regioselectivity in KT2440** (C-8 vs C-6) should be verified by product identification, as some biotransformations yield 6- and 8-hydroxy products.
 - A predicted structure exists (AlphaFoldDB Q88I03); docking naringenin into the active site and mapping the FAD/NAD(P)H pockets could pinpoint substrate-determining residues.
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10. Key References

- Marin et al. 2016, *Appl Environ Microbiol* — PMID **27059806** (FdeE monooxygenase catalyzes first step; naringenin → 8-hydroxynaringenin; FdeC A-ring cleavage → oxaloacetate/TCA).

- Dulak et al. 2024 — PMID **39107441** (purified FdeE = regioselective flavonoid **C-8 hydroxylase (F8H)** FMO; broad flavonoid substrate range).
- Wassem et al. 2017 — PMID **27878922** (NodD-like LysR regulator FdeR; flavonoid-inducible `fde` operon).
- Marin et al. 2013 — PMID **23125118** (identification of the `fde` operon; `fdeA` / `fdeR` required for naringenin degradation).
- Supporting database evidence: UniProt Q88I03; KEGG ppu:PP_3199 and neighbors; InterPro IPR002938/IPR050493/IPR036188; Pfam PF01494.

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